ISO 9001
ISO 14001
OHSAS 18001

swam

• Technology • Innovation • Excellence

Liquid Ring Vacuum Pumps for Power Industry

......the next generation technology

Unmatched Technology,
Superior Performance,
Trouble Free Operation

THE COMPANY

Swam a group known for technology, innovation and excellence is the world's largest manufacturers of Pressure & Vacuum Technology Blowers and Pumps.

Swam Liquid Ring Technology, developed by our technical, highly skilled engineers are of advanced design, superior performance and of high reliability. The design and construction of the Liquid Ring Vacuum Pump and Compressors reflect our experience and know-how acquired over years of product development in this particular field of machine engineering. These pumps are used in industries like Paper, Sugar, Coal, Chemical, Steel Plants, Pharmaceuticals, Fertilizer, Food Processing, Asbestos, Mines, Solvent Extractions, Washeries, Powerplants, Oil Refineries, Textile & Tyre Industries, Distilleries, Railways, etc.

With over three decades of experience, the company has adequately matured to execute different projects and is today represented in over 15 countries. Our quality system has been accredited with ISO-9001 Quality Standards and procedures. The company's head office and works are located at Noida near New Delhi, the capital city of India. The company has FOUR manufacturing plants equipped with CNC Machines and in-house testing facility upto 1000 kw.

Our highly motivated engineers and work force faces, each contract with commitment to meet the client process specification & quality requirement. The company has executed many prestigious projects under the stringent quality compliance.

The company's quality system is as per ISO 9001 : 2008 and has very high regard for safety and environment and hence has it's all plants ISO 14001 : 2004 and OHSAS 18001 : 2007 certified.

ADVANCE MANUFACTURING SYSTEM

Swam has the most advance manufacturing system with a large number of CNC Machines installed in our four plants. The Liquid Ring Vacuum Pumps and other products are manufactured with most optimum profile resulting higher efficiencies, high displacement, lower slippage. Thus our products are more energy efficient compared to others.

SWAM - LIQUID RING VACUUM PUMPS FOR POWER INDUSTRY

Condenser Exhauster Pumps

Swam offers advanced design Condenser Exhauster Pumps for hogging and holding service in power generation. The system consists of liquid ring vacuum pump, separator, heat exchanger and ejector (optional) with total recirculation.

Capacity : As per requirement

Fly Ash Conveying

Swam offers advanced design Liquid Ring Vacuum Pumps for pneumatic conveying of fly ash in the Power Plants under negative pressure i.e. vacuum services. They are of advanced design and can handle high quantity of ash.

Capacity : Upto 36000 M^3/ Hr. to 90% Vacuum.

Flue gas desulphurisation Plants

Flue Gas Desulphurisation system are a common feature for ultra modern coal fired power plants, a by-product of the process is gypsum which is produced at the outlet of a scrubber. Liquid Ring Vacuum Pumps are used to provide vacuum for de-watering of the gypsum on a vacuum filter, usually belt or rotary drum type. The Liquid Ring Pump, typically a large single stage design, is selected based upon the surface area of the vacuum filter.

Water Box Priming System

The ability to handle wet gases without detrimental effect makes Liquid Ring Vacuum Pumps an ideal choice for the priming application. The pumps are used initially to carry out priming of the main condenser water boxes and CW pump; once this is complete, they are used to maintain vacuum in the water box at the required level.

LATEST RELIABLE TECHNOLOGY FOR THE INDUSTRIAL PROCESSES

Swam pumps provide lower maintenance & lower life cycle costs and greater overall value to our clients. The Liquid Ring Vacuum Pumps have improved design, enabling wider range of operation, higher efficiency, saving of water and simpler seal line piping.

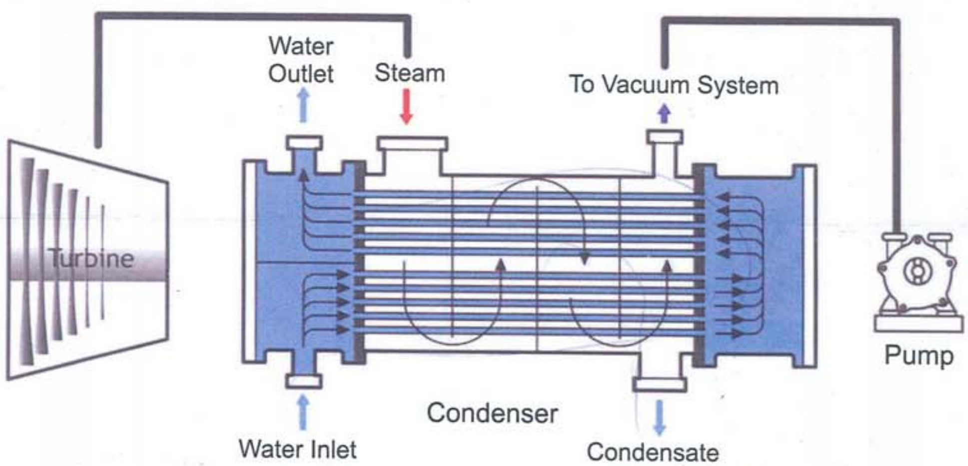
CONDENSER AIR EXTRACTION

It has been proven and known fact that to increase the operational performance and reliability of turbine condenser vacuum plants, the need is to focus on the dynamic relationship between the vacuum pumping system, condenser performance and turbine back pressure. This is an important factor in the design requirements and needs careful appraisal when considering overall plant performance.

Swam condenser air extraction packages, are based on two-stage Liquid Ring Vacuum Pumps, designed to remove system air leakage across common sizes of turbine generator steam condensers. This is achieved by reducing the pressure in the turbine, enabling more heat from the steam to be converted to mechanical energy, increasing efficiency of the power plant.

The air load from the turbine condenser system is saturated with vapour : Liquid Ring Pumps are ideal for handling high vapour loads as much of the vapour will be condensed at the pump suction (by the direct condensing action of the inlet water spray or contact with the pump seal water). This condensing reduces the total volume to be handled by the pump, which is a significant advantage when compared to other pumping technologies.

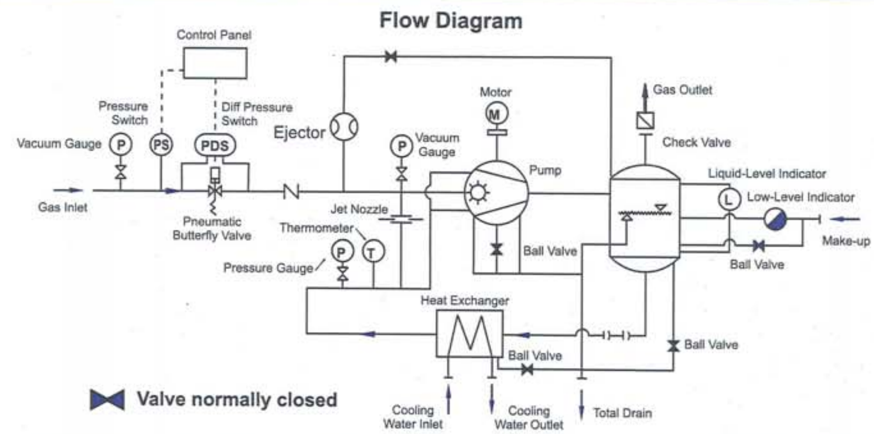
Thus for the right selection of the vacuum equipment it is usually based upon theoretical condition, with performance requirements taken from HEI (or VGB) recommendations. Client specifications require further analysis of pump performance at actual operating temperatures & pressures, including part load and operation with alternate fuels.



Swam not only provides vacuum pump packages and reliable service, but also make contributions to the renewal of existing stations.

In the technical improvement process of power plants, using Swam corresponding pump sizes can make the operation more reliable and improve the efficiency of units.

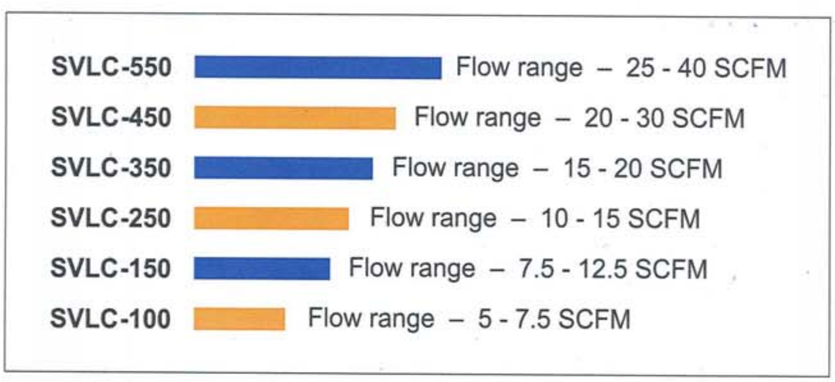
Typical schematic P&ID of a Condenser Exhaustion Vacuum Pump Package



SWAM - Standard Packages & Sizes As Per HEI Standard

Swam standard packages are designed to meet the requirements of HEI standards, Initial temperature difference is an important consideration in optimum pump selection. The pump sizing and selection is based upon varying ITD's at 1" hg abs (33.86 mbar'a) at the condenser outlet.

Air Flow Capacity of SWAM - SVLC - Series Liquid Ring Vacuum Pump



WATER BOX PRIMING

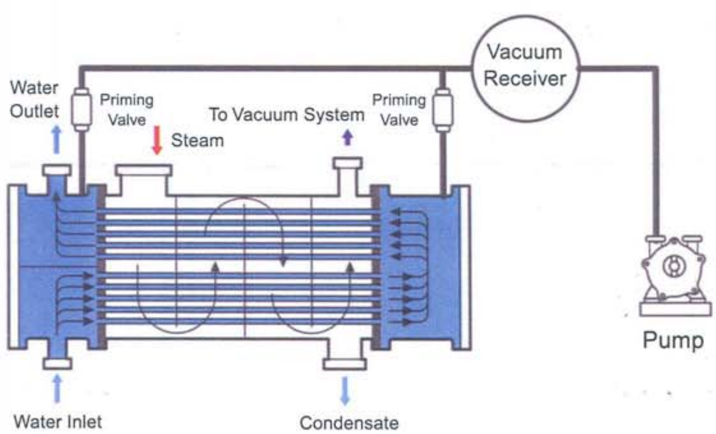
It is due to the advantage of Liquid Ring Pumps ability to handle wet gases without detrimental effect makes Liquid Ring Vacuum Pumps ideal for the priming application. The pumps are used initially to carry out priming of the main condenser water boxes and CW pump; once this is complete they are used to maintain vacuum in the water box at the required level.

The removal of air from the condenser water box prevents accumulation of air in the upper parts of the cooling tube bundle, thereby preventing air locks and maintaining the effective cooling surface of the condenser, ensuring maximum cooling efficiency.

The **Swam** priming system is a package design consisting of three parts: Vacuum Pump, Vacuum Receiver and Priming Valve and accessories, which are supplied as required to meet customer specifications.

The vacuum pump system consists of a single stage Liquid Ring Pump complete with a total seal water recirculation system, sized to meet the process duty requirements. If duty & standby pump sets are called for, then two systems are provided, giving true standby capability including the seal water recirculation system.

The vacuum receiver vessel is complete with a pressure transmitter to control the pump operation & includes an automatic drain arrangement to remove any water carry over from the system, therefore avoiding corrosion. If required, a priming valve and associated accessories can be supplied for each condenser vacuum connection: the priming valve helps to prevent cycling of the vacuum system and minimises carry over of cooling water.



ASH CONVEYING APPLICATIONS

Swam Liquid Ring Pumps incorporate state-of-the-art design for pneumatic conveying of ash in negative pressure (vacuum) in Power Plants. This application consists of pneumatically conveying fly ash from the precipitation hoppers to a central dry collection point under vacuum. The main advantage of this type of conveying system is that since the ash is being conveyed under vacuum, any leakage is of air inward & not ash outward. For this the pumps are available upto very large sizes of 30,000 m^3/hr. and are highly reliable. The maintenance of the pumps is negligible and performance is excellent.

FLUE GAS DESULPHURISATION

Flue Gas Desulphurisation system are a common feature for ultra modern coal fired power plants in which gypsum is a by-product of the process and is produced at the outlet of a scrubber. Liquid Ring Pumps are used to provide vacuum for the dewatering of the gypsum on a vacuum filter, which is usually belt or rotary drum type. The Liquid Ring Pump, typically a large single stage design, is selected based upon the surface area of the vacuum filter.

TESTING & QUALITY SYSTEMS

Swam is ISO 9001, ISO 14001 and OHSAS 18001 certified company and therefore quality and safety is the first thing to maintain. Besides this all our products comply to International Standards and are CE certified. The quality control is practiced at all levels in the organization and in general the quality is maintained through the established quality manuals and plans. This also follows the quality control manual and procedures drawn out for ISO-9001 quality certification.

Swam has in-house testing facility for testing Blowers, Compressors, Vacuum Pumps, Centrifugal Fans etc. with a number of test beds. Each product made to run for several hours as per the standard and codes before taken for painting & dispatch i.e. The company has in-house testing facility upto 1000 kw. and for 415 v, 690 v, 3.3 kv., 6.6 kv. varying voltage range.

FOCUS ON QUALITY & RELIABILITY

Emphasis on quality and performance is a commitment of every individual at **Swam** beginning at grass root level. The quality control is practiced at all levels of the organization and all its aspects are thoroughly complied with the specified standard & specification, till the product gets delivered.

The quality assurance system encompasses self quality control by each worker, supervised by quality control engineers. Each product undergoes thorough and stringent tests at different stages of manufacturing & are test run for several hours prior to dispatch as per the relevant code, customers' specification and approved quality plan.

These functions are carried out by our dedicated team of qualified experienced engineers.

ISO 9001
ISO 14001
OHSAS 18001

Customer Support Services:

A team of experienced and qualified engineers are deployed to rendering all kind of technical post supply services to our esteemed customers; such as pre-commissioning and post commissioning services, trouble shooting etc., to make the equipment perform to user's/ clients satisfaction.

Professionally and technically competent personnel are always ready to render these services, promptly and expeditiously.

For details contact H.O. or the nearest service network.

Applications:

- Oil and Gas Plants
- Pulp & Paper Industries
- Power Plants
  - Ash Conveying
  - Negative Conveying
  - Condenser Exhauster
- Filter Plants
- Sugar Industries
- Process Industries
- Mineral Beneficiation Plants / Coal Washeries
- Chemical Plants
- Ship Building
- Textile Industries
- Petrochemical Plants
- Jet and Surface Condensers and various other uses

Manufacturing Plants:

swam

Technology • Innovation • Excellence

SWAM PNEUMATICS PVT. LTD.

(ISO - 9001:2008 Quality Certified)

Corporate Head Office & Plant-I :
C-2, Sector-3, Noida-201301, Gautam Budh Nagar (INDIA)
Tel : +91-120-4696222, Fax : +91-120-2443283, 4696200
E-mail : sales@swamatics.com

Plant-II : B-9, Sector-57, Noida-201301 (INDIA)
Plant-III : B-111, Sector-67, Noida-201307 (INDIA)
Plant-IV : C-52, Sector-57, Noida-201301 (INDIA)

Website : www.swamatics.com

For details of our local contact office in your region, contact H.O. or log on to website

ISO 9001 URS ISO 9001 ISO 9001 ISO 9001 UKAS QUALITY MANAGEMENT

Leader in : Air-Gas Conveying , Boosting & Vacuum Technology